Bryan Tseng
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EDUCATION

Johns Hopkins University School of Medicine, Baltimore, MD **August 2024 - Current**
Ph.D. in Biomedical Engineering Advised by Dr. Adam S. Charles
Activities: **Treasurer** for Biomedical Engineering Ph.D. student council

Boston University College of Arts and Sciences, Boston, MA **September 2016 - May 2020**
Bachelor of Science in Mathematics (Statistics) and Biology (CMG)
Minor in Economics
Awards: Dean's List (all-semesters), UROP research award (Fall 2019 and Spring 2020), magna cum laude

PUBLICATIONS

Peer-reviewed:

1. Guay C, Agrawal U, Tseng B, Gallo S, Schreier D, Brown E (2025). Clinical Electroencephalography for Anesthesiologists and Intensivists Part II: Physiologic Signatures and Active Management, Anesthesiology.
2. Breault M, Orguc S, Kwon O, Kang G, Tseng B, Gallo S, Schreier D, Brown E (2024). Anesthetics As Treatments For Depression: Clinical Insights and Underlying Mechanisms, Annual Review of Neuroscience
3. Subramanian S, Tseng B, Carmen M, Goodman A, Dahl D, Barbieri R, Brown E (2024). Monitoring Surgical Nociception Using Multimodal Physiologic Markers, Proceedings of National Academy of Sciences.
4. Subramanian S, Tseng B, Barbieri R, Brown E (2022). An unsupervised automated paradigm for artifact removal from electrodermal activity in an uncontrolled clinical setting, Physiological Measurement.
5. Marshall A, Joyce C, Tseng B, Gerlowvin H, Yeh G, Sherman K, Saper R, and Roseen E (2021). Changes in Pain Self Efficacy, Coping Skills and Fear Avoidance Beliefs in a Randomized Controlled Trial of Yoga, Physical Therapy, and Education for Chronic Low Back Pain, Pain Medicine
6. Nojiri T., Chen CY., Kim DM, Silva DA, Lee C, Maeno M, McClelland AA, Tseng B, Nagai S, Hatakeyama W, Kondo H, and Nagai M (2019). Establishment of perpendicular protrusion of type I collagen on TiO2 nanotube surface as a priming site of peri-implant connective fibers, Journal of Nanobiotechnology 17(1):34

Conference Papers:

1. Tseng B, Subramanian S, Barbieri R, Brown E (2022). Tonic Electrodermal Activity is a Robust Marker of Psychological and Physiological Changes during Induction of Anesthesia, Proc. 44th IEEE International Conf on Eng in Biol and Med (EMBC). Glasgow, UK, Jul.
2. Subramanian S, Tseng B, Barbieri R, Brown E (2021). Unsupervised Machine Learning Methods for Artifact Removal in Electrodermal Activity, Proc. 43rd IEEE International Conf on Eng in Biol and Med (EMBC). Virtual, Oct 31st - Nov 4th

Conference Abstracts:

1. Tseng B, Subramanian S, Purdon P, Barbieri R, Brown E (2021). Prediction of unconsciousness using clinically available markers. 43rd IEEE International Conf on Eng in Biol and Med (EMBC). Virtual, Oct 31st - Nov 4th
2. Tseng B, Subramanian S, Purdon P, Barbieri R, Brown E (2021). Multimodal brain-heart analysis reveals subject-specific dynamics during propofol anesthesia. Organization for Human Brain Mapping (OHBM). Virtual, Jun 21st - Jun 25th

Provisional Patents:

1. Combinations of Anesthetics as Therapies for Treatment-Resistant Depression (MIT 25342)

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, MD **August 2020 - May 2024**
Department of Biomedical Engineering
Graduate Research Assistant, with Dr. Adam S. Charles

- **Research topic:** Developing **adaptive stimulation algorithms** for brain-computer interfaces by learning interpretable models of **neural dynamics** and predictive models of movement.
- Designed reaching and head-turn behavioral paradigms to study the effect of **microstimulation** in the deep cerebellar nuclei using **Neuropixel** Read Write (NPRW) prototype probes
    - Assisted with the non-human primate neural recording
- Developed the analysis pipeline based on SpikeInterface for simultaneous high-density electrophysiology recording from **Utah arrays**, NPRW probes, and **DeepLabCut** annotated video kinematics data

    - Neural recording analysis pipeline included artifact correction of stimulation artifact, spike detection, and firing rate estimation
- Analyzed the parameter space of NPRW microstimulation and its corresponding effects on behavior kinematics and evoked neural responses (cortical and deep brain)
- Analyzed behaviorally relevant subspaces of neural dynamics in the deep cerebellar nuclei
- Deployed analysis pipelines on the high-performance computing cluster provided by Center of Imaging Science

Massachusetts Institute of Technology, Cambridge, MA **August 2020 - May 2024**
Neuroscience Statistics Research Laboratory (NSRL), Massachusetts General Hospital
Technical Associate, with Dr. Emery N. Brown

- Implemented **state-space multitaper spectral analysis** on the electroencephalography (EEG) to track patient's level of consciousness during general anesthesia
    - Assisted in integrating the online EEG metrics into an existing **closed-loop anesthesia delivery** system
- Implemented **point-process** and **state-space methods** to track patient's nociceptive state during general anesthesia
    - Derived from a history-dependent Inverse Gaussian model on the electrocardiogram (EKG) and electrodermal activity (EDA), reaching accuracies of up to 75%
- Investigated the neurophysiological mechanisms underlying the **antidepressant effects of anesthetics**
- Investigated temporal dynamics of brain activity, autonomics, and behavior due to propofol administration
- Standardized the lab database and analysis pipeline and integration with the MIT OpenMind computing cluster
- Fascilitated the collaboration between NSRL and the Barbieri lab at Politecnico di Milano

Clinical responsibilities at Massachusetts General Hospital:

- Led and initiated passive data collection efforts of physiological signals in the ICU, operating room, and electroconvulsive therapy units
    - Physiological signals from 100+ patients (EEG, EKG, EDA, photoplethysmography)
- Performed **artifact correction** and peak extraction of physiological signals using **machine learning** methods
    - Tested isolation forest, K-nearest neighbor distance 1-class support vector machine methods to remove cautery artifact in data collected during surgery
- Investigated the clipping and scaling issues of EEG signals from the Masimo Sedline Root device
- Managed IRB protocols and data use agreements at MGH

Boston Medical Center, Boston, MA **June 2019 – August 2020**
Research Assistant, with Dr. Robert Saper and Eric Roseen

- Investigated psychometric effects of yoga and physical therapy on adults with chronic lower back pain
    - Assessed within and between-group change scores of covariates from the Back to Health study in SAS
    - Implemented **stochastic regression imputation** of missing data
- Investigated attitudes and beliefs of opioid use disorder patients on non-pharmacologic treatment
- Performed **Fisher exact analysis** on the effects of nonpharmacologic treatment relative to pain medication usage
- Funded by the Undergraduate Research Opportunities Program (UROP) (Fall 2019 and Spring 2020)

LEADERSHIP EXPERIENCE

Allocation Board of Boston University, Boston, MA **September 2017 – May 2020**
Chair

- Worked directly with the Dean of Students to allocate over $700,000 annually to 300 clubs
- Led weekly meetings to approve funding requests from student organizations
- Worked with club presidents and treasurers to ensure the proper use of funding
- Rewrote the board constitution, handbook, performed audits, and helped design a new recruitment process
- Facilitated the transition to a new finance and event planning platform through workshops and advertisement
- Interviewed and led training processes for new members
- Helped to coordinate large campus-wide events such as the first BU spring concert

VOLUNTEER EXPERIENCE

- **Peer mentor** at Boston University Department of Mathematics and Statistics **September 2019 - May 2020**
    - One-on-one mentorship for Freshman students, hosted department-sponsored events
- **Big Brothers and Big Sisters of Mass Bay** **September 2017 - May 2019**
    - Mentorship of youth in the Greater Boston Area in a biweekly camp